Understanding the Problem

How did you approach understanding the challenge?

I first read through the exercise and identified what the program was required to have. I then broke the problem into smaller parts, input, process, and output. In each section I worked to optimize the code, and I added extra features that weren't required, such as error handling. I spent the majority of my time refining the masteries, as they matter more than the skilbuilders.

Were there any parts of the problem you found confusing at first? If so, how did you resolve that confusion?

I initially found the calculations difficult, and how to make the program calculate prime numbers, and perform other operations. I resolved this by familiarizing myself with the different codes for performing calculations

Planning the Solution

Did you create a plan or break the problem into smaller steps before coding?

Yes I broke the program down into steps. First I planned how to get the user's input, then I planned what calculations the computer would perform with that information, and finally I planned what the computer would output.

How did you decide on the tools, data structures, or algorithms to use?

I used the tools that I believed were best for the given task. I used loops for error handling, booleans for true or false statements, and simple integers to perform calculations.

Implementation

Did you write the code in small pieces or attempt the entire solution at once?

I had already planned out each section of the code in smaller pieces, so when I wrote it I took inspiration from my planning and worked on each section individually while ensuring that they all worked together in harmony.

How did you test your solution along the way to make sure it was working?

I would run the program many times throughout the coding process. I wanted to ensure that each new line of code performed its task correctly before continuing.

Overcoming Challenges

What part of the problem was the most difficult for you?

For me the most challenging part was generating every prime number between the user inputted numbers. I solved this simply by creating a loop to check every single number in between the two, and only print the ones that are prime.

How did you handle moments when you felt stuck or unsure of what to do next?

I looked at the code carefully, examined the error messages, and conducted research to learn why it wasn't performing its task correctly and how I could fix it. I also looked back on my previous code to see what I was doing differently.

Learning

Was there anything you learned that you think will help you with future challenges?

I learned how much is truly possible with Java through its math function. There are calculations that can be performed for nearly every task, such as square roots, prime numbers, etc.